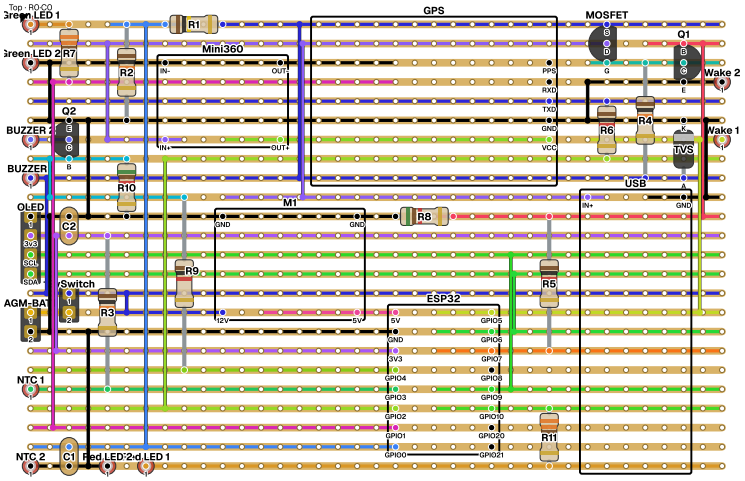
Bike-Mate V1.10 | placement (component side)

Print at 100% / actual size, disable "fit to page", then verify the ruler. Lay this on the component side and push leads through the marked holes.

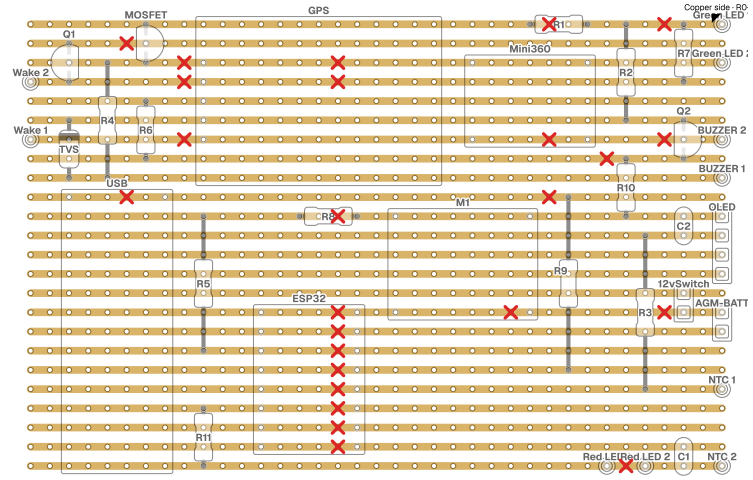
50 mm - measure this. If it isn't exactly 50 mm, reprint at 100% (no "fit to page").



Bike-Mate V1.10 | track cuts (copper side, mirrored)

Mirrored for the underside. Cut each ×. Verify the ruler.

50 mm - measure this. If it isn't exactly 50 mm, reprint at 100% (no "fit to page").



Bill of materials

Qty	Component	Value	References
2	Capacitor	100nf	C1, C2
7	Connector (2-pin)	-	12vSwitch, AGM-BATT, BUZZER, Green LED, NTC, Red LED, Wake
1	Connector (4-pin)	-	OLED
1	Diode	SA16CA	TVS
1	ESP32-C3 SuperMini	-	ESP32
1	Mini360	-	Mini360
1	MP1584EN	-	M1
1	NEO-6M GPS	-	GPS
2	NPN Transistor	-	Q1, Q2
1	P-Channel MOSFET	-	MOSFET

Qty	Component	Value	References
1	Resistor	100k	R1
2	Resistor	10k	R2, R3
1	Resistor	10K	R4
3	Resistor	1K	R5, R6, R9
2	Resistor	330	R7, R11
2	Resistor	5K1	R8, R10
1	USB Charger	-	USB

Bike-Mate V1.10 | off-board wiring

These parts are mounted off the board. Run one wire from each pin to its solder pad, which is labelled on the component sheet.

Part	Pin	Net	Solder pad
BUZZER, Connector (2-pin)	1	net18	row 9, column 1
	2	net8	row 7, column 1
NTC, Connector (2-pin)	1	net5	row 20, column 1
	2	GND	row 24, column 1
Green LED, Connector (2-pin)	1	net14	row 1, column 1
	2	GND	row 3, column 1
Red LED, Connector (2-pin)	1	net16	row 24, column 7
	2	GND	row 24, column 5
Wake, Connector (2-pin)	1	net12	row 7, column 37
	2	GND	row 4, column 37